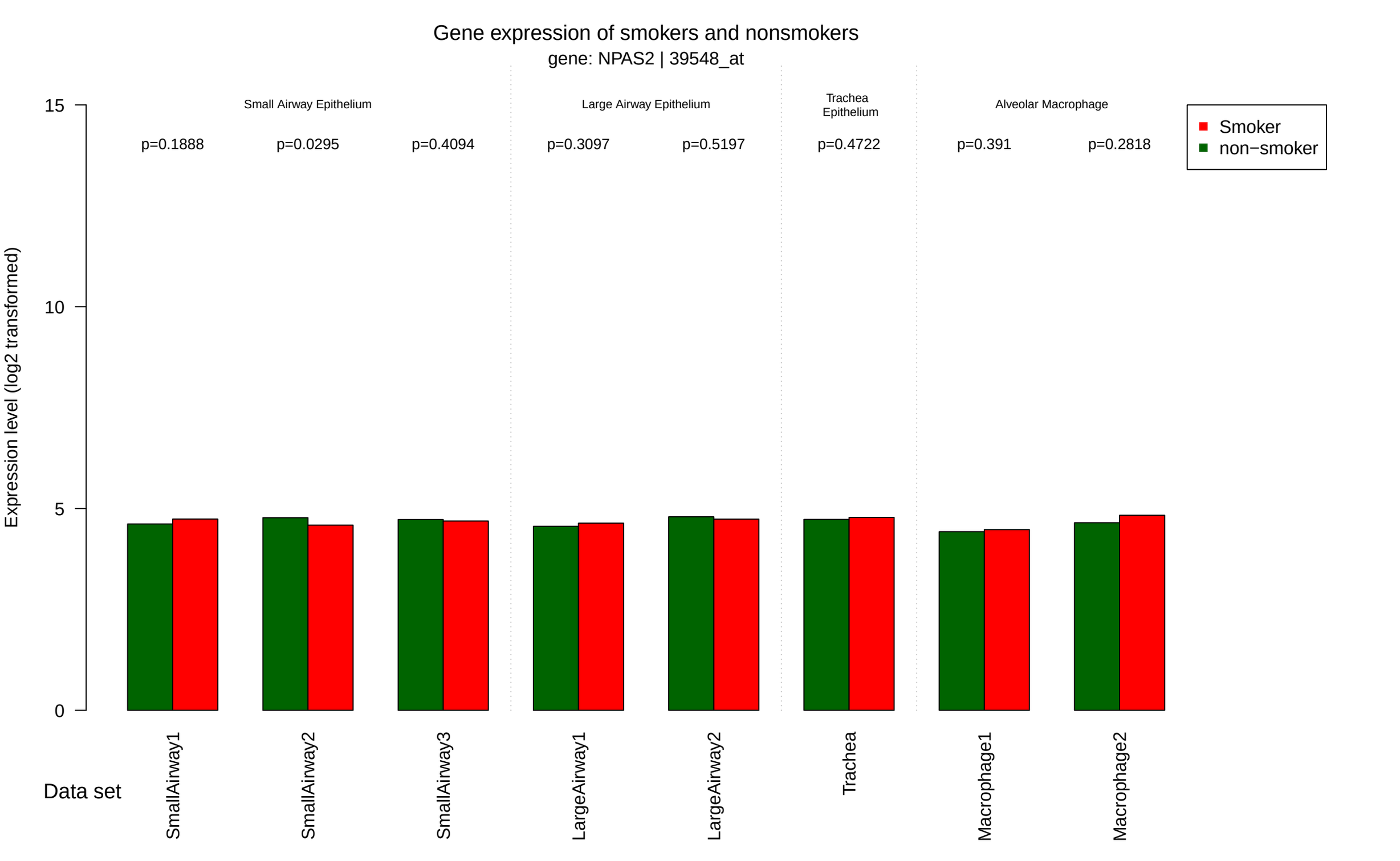
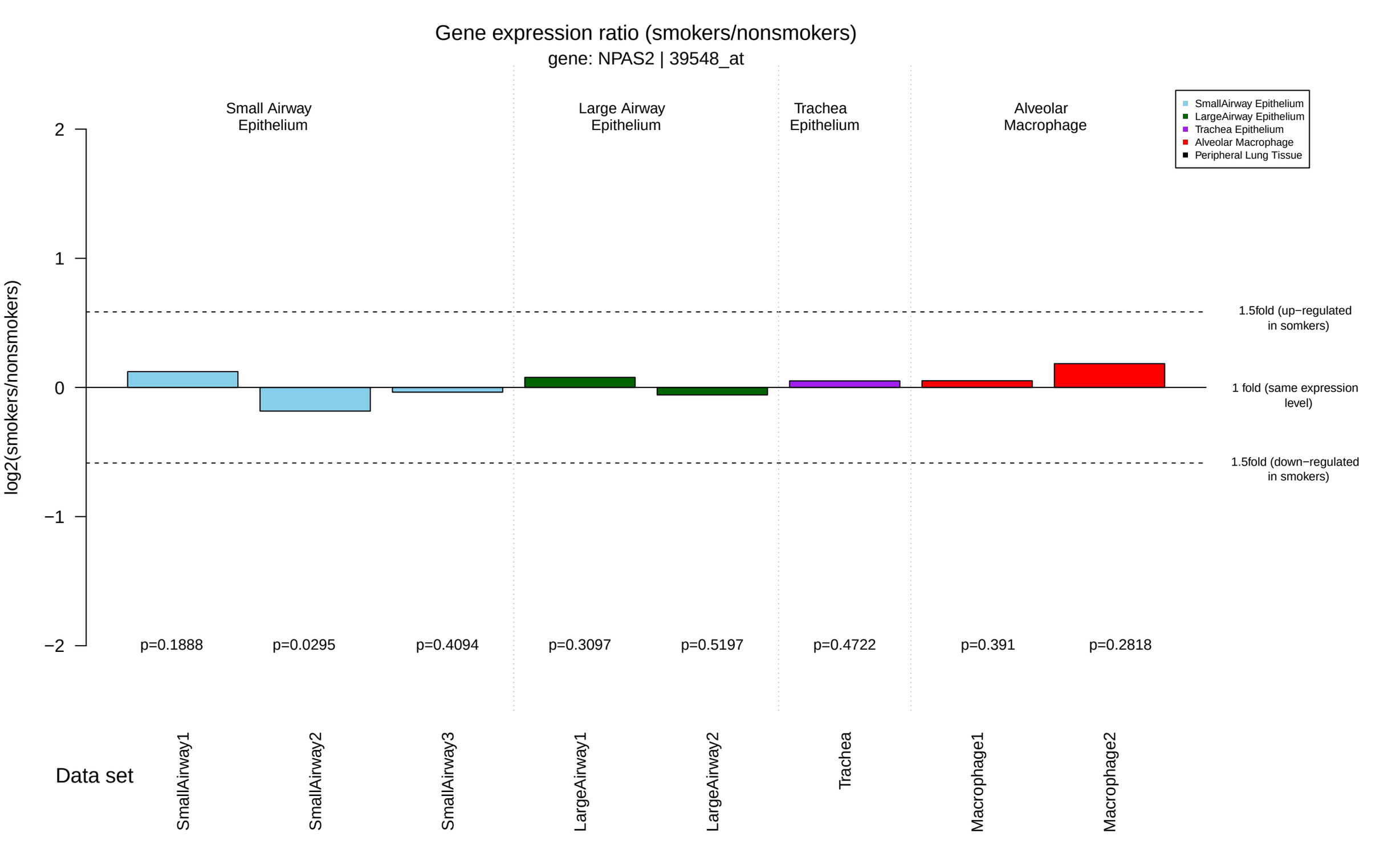
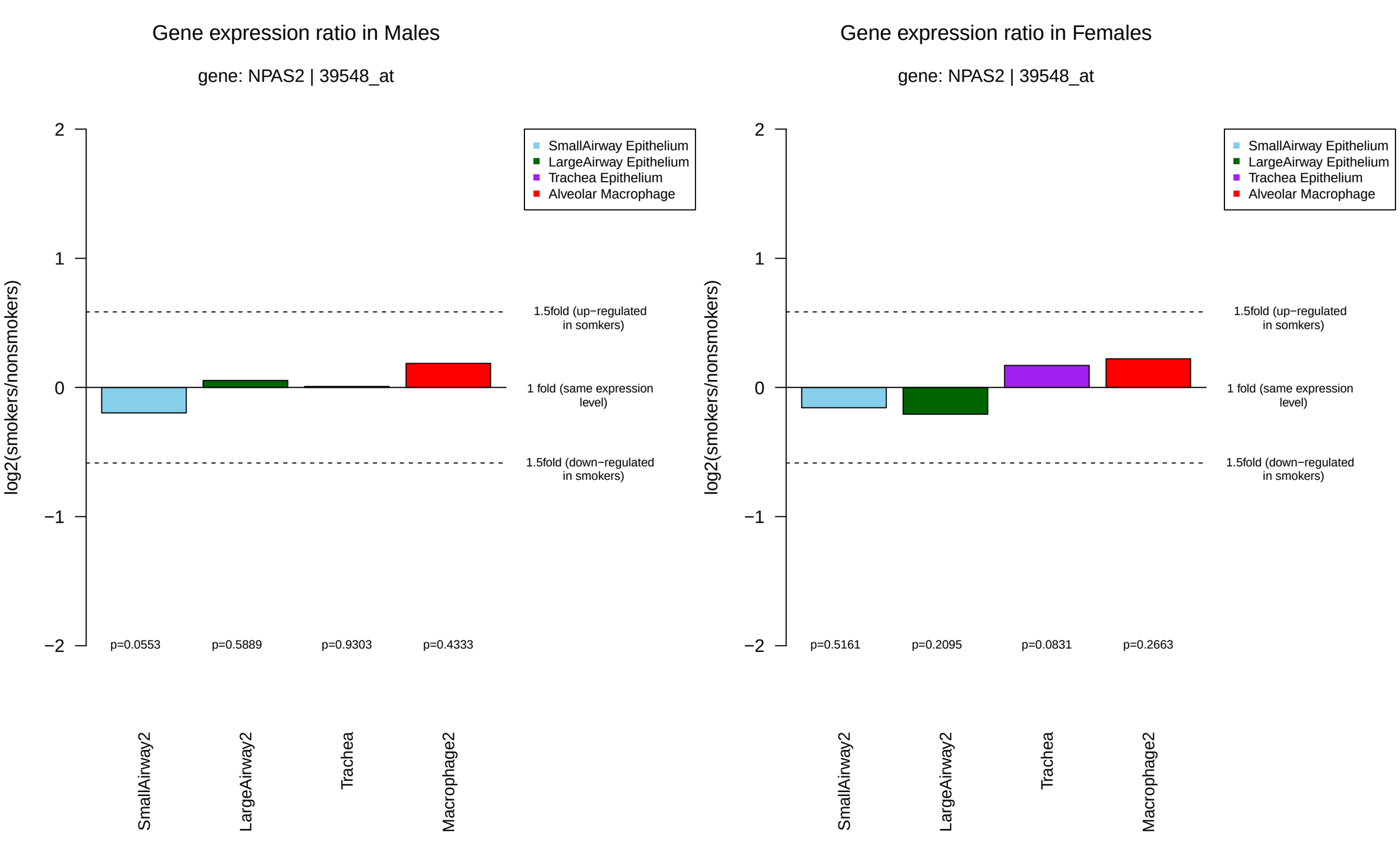
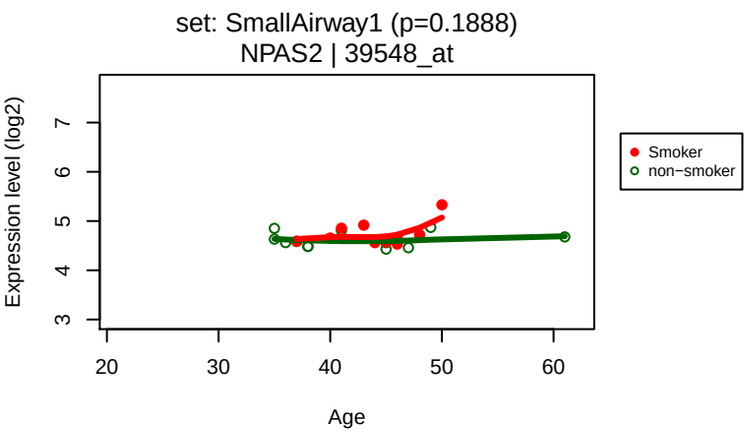
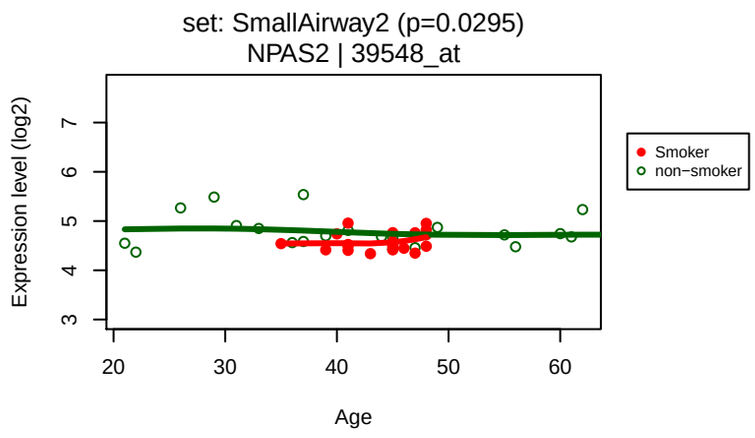
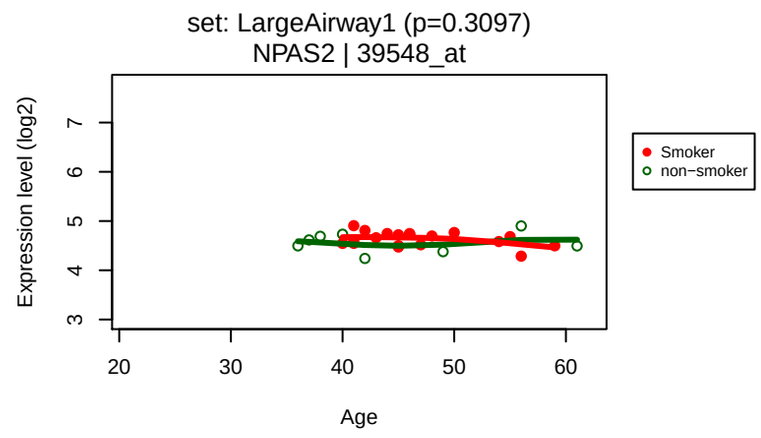

Gene expression of smokers and nonsmokers
gene: NPAS2 | 39548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1888
p=0.0295
p=0.4094
p=0.3097
p=0.5197
p=0.4722
p=0.391
p=0.2818
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

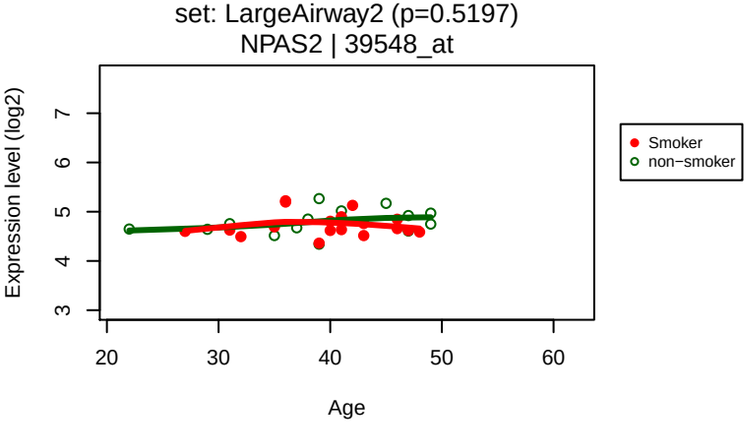

Gene expression ratio (smokers/nonsmokers)
gene: NPAS2 | 39548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1888
p=0.0295
p=0.4094
p=0.3097
p=0.5197
p=0.4722
p=0.391
p=0.2818
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

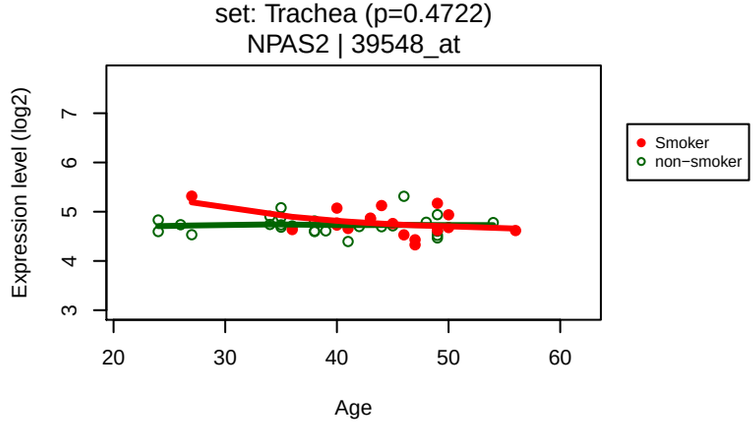

Gene expression ratio in Males
gene: NPAS2 | 39548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0553
p=0.5889
p=0.9303
p=0.4333
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPAS2 | 39548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5161
p=0.2095
p=0.0831
p=0.2663
SmallAirway2
LargeAirway2
Trachea
Macrophage2

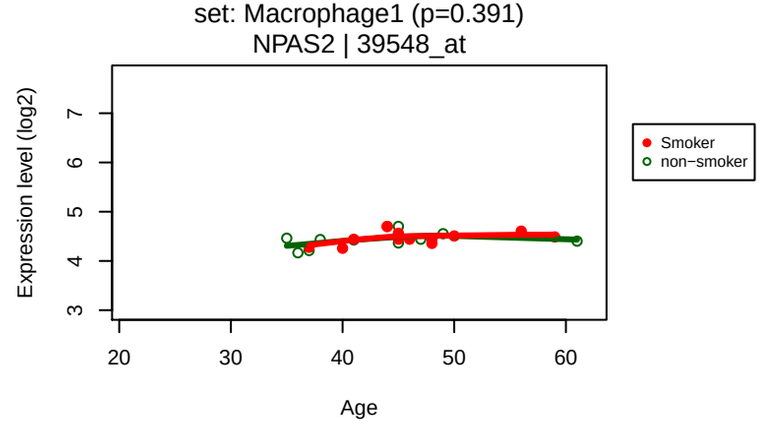

set: SmallAirway1 (p=0.1888)
NPAS2 | 39548_at
Expression level (log2)
Age
Smoker
non−smoker

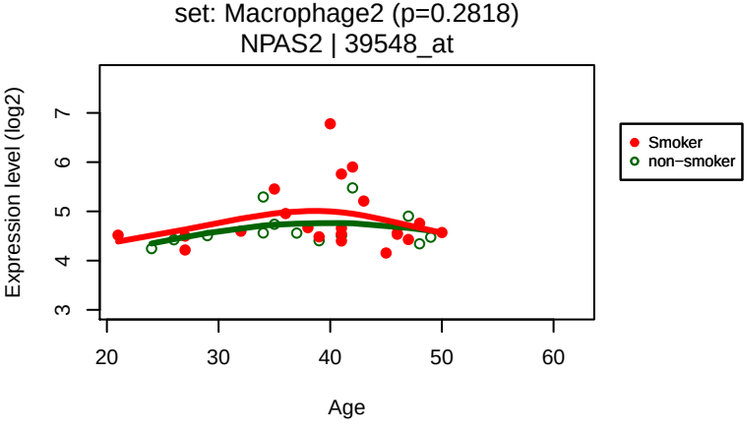

set: SmallAirway2 (p=0.0295)
NPAS2 | 39548_at
Expression level (log2)
Age
Smoker
non−smoker

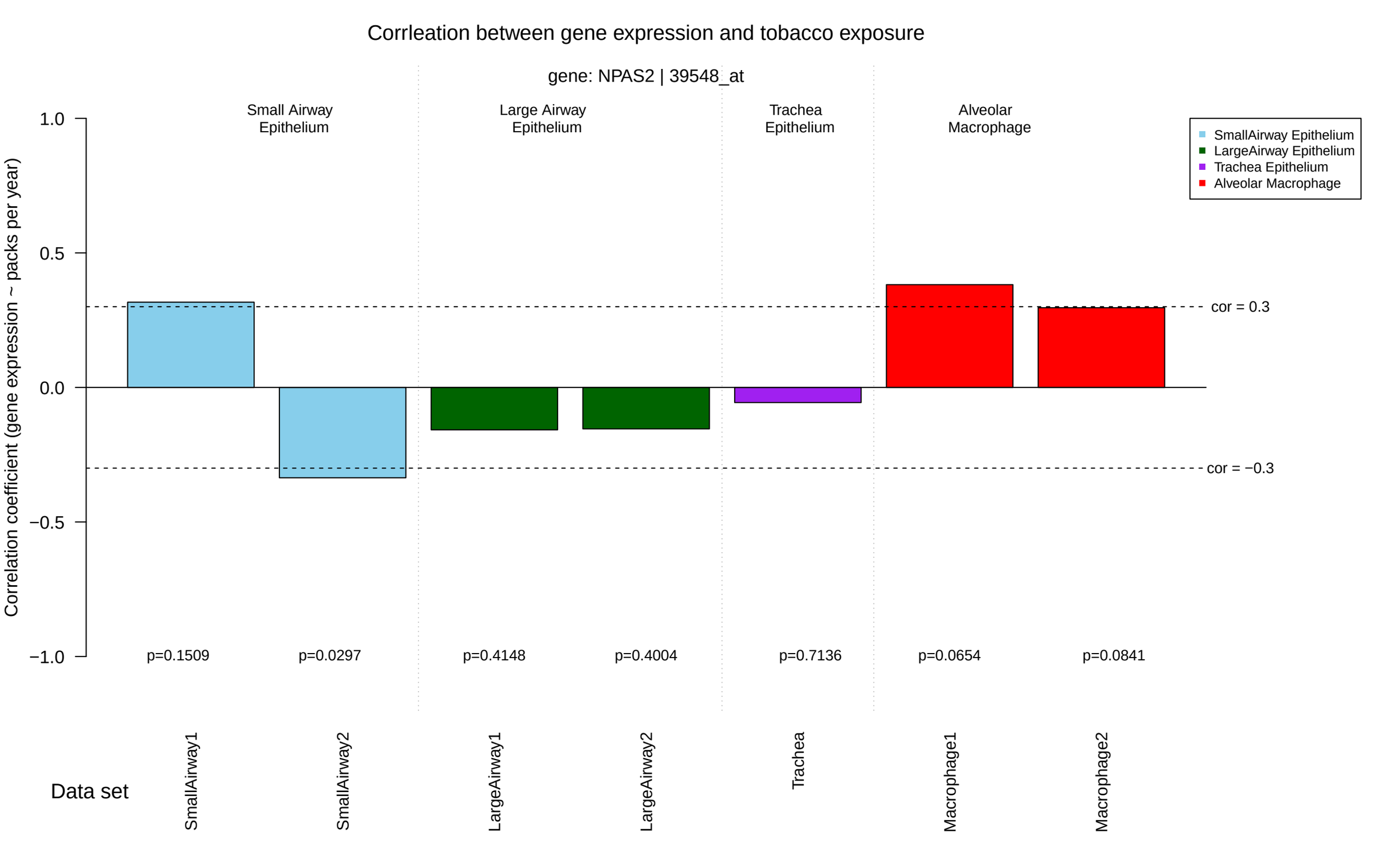

set: LargeAirway1 (p=0.3097)
NPAS2 | 39548_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5197)
NPAS2 | 39548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4722)
NPAS2 | 39548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.391)
NPAS2 | 39548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2818)
NPAS2 | 39548_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPAS2 | 39548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1509
p=0.0297
p=0.4148
p=0.4004
p=0.7136
p=0.0654
p=0.0841
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2